# Chapter 1

# Gaussian integrals and Hermite polynomials

The paper's first theorem says that the Ornstein–Uhlenbeck channel multiplies the Hermite moment of degree $m$ by $e^{-mt}$, exactly, for every law with finite moments of that order. Its proof is four lines long once three classical facts are in place: the Gaussian generating function, the generating function of the Hermite polynomials, and Mehler's formula. This chapter builds those facts from the Gaussian integral upwards, proves Mehler's formula and the paper's Theorem 1 in full, derives the two exact consequences the paper states after it, works out the Wick (Isserlis) rule and the second-moment lemma that produce the constants $\frac{1}{6}$ and $\frac{1}{24}$, explains when a density ratio belongs to $L^2(\gamma)$ and why the paper does not need it to, and defines the first Edgeworth correction that the chapter on misspecification uses.

## 1.1 Gaussian integrals

**Lemma 1.1** (The scalar Gaussian integral and its moments)**.** $\int_{\mathbb{R}} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$. *Writing* $\gamma(x) = (2\pi)^{-1/2}e^{-x^2/2}$ *for the* ***standard Gaussian density*** *and* $Z \sim \gamma$,

$$\mathbb{E}[e^{sZ}] = e^{s^2/2}, \qquad \mathbb{E}[Z^{2k}] = (2k-1)!! = \frac{(2k)!}{2^k k!}, \qquad \mathbb{E}[Z^{2k+1}] = 0. \tag{1.1}$$

*Proof.* Squaring the integral and passing to polar coordinates,

$$\Big(\int e^{-x^2/2}\,\mathrm{d}x\Big)^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\;\mathrm{d}r\;\mathrm{d}\phi = 2\pi.$$

Completing the square, $sz - z^2/2 = -(z-s)^2/2 + s^2/2$, so

$$\mathbb{E}[e^{sZ}] = e^{s^2/2}(2\pi)^{-1/2}\int e^{-(z-s)^2/2}\,\mathrm{d}z = e^{s^2/2}.$$

Expanding both sides in $s$, $\sum_n \mathbb{E}[Z^n]s^n/n! = \sum_k s^{2k}/(2^k k!)$, and comparing coefficients of $s^{2k}$ gives $\mathbb{E}[Z^{2k}] = (2k)!/(2^k k!)$, which equals $(2k-1)!! = 1\cdot 3\cdots(2k-1)$; odd coefficients vanish. The exchange of sum and expectation is justified by $\sum_n |s|^n|Z|^n/n! = e^{|s||Z|}$, which is integrable. □

So $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, $\mathbb{E}Z^6 = 15$. The number $(2k-1)!!$ is the number of ways of pairing $2k$ objects, which is the content of the next section.

**Lemma 1.2** (The multivariate Gaussian integral)**.** *For a symmetric positive definite* $\mathbf{A}$ *on* $\mathbb{R}^L$ *and* $\mathbf{b} \in \mathbb{R}^L$,

$$\int_{\mathbb{R}^L} \exp\!\big(-\tfrac{1}{2}\mathbf{x}^{\top}\mathbf{A}\mathbf{x} + \mathbf{b}^{\top}\mathbf{x}\big)\,\mathrm{d}\mathbf{x} = (2\pi)^{L/2}(\det \mathbf{A})^{-1/2}\exp\!\big(\tfrac{1}{2}\mathbf{b}^{\top}\mathbf{A}^{-1}\mathbf{b}\big). \tag{1.2}$$

*Consequently $\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})$ has density $(2\pi)^{-L/2}(\det \mathbf{\Sigma})^{-1/2} \exp(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^\top \mathbf{\Sigma}^{-1}(\mathbf{x}-\mathbf{m}))$ and generating function $\mathbb{E}[e^{\mathbf{s}^\top \mathbf{X}}] = \exp(\mathbf{s}^\top \mathbf{m} + \frac{1}{2}\mathbf{s}^\top \mathbf{\Sigma} \mathbf{s})$.*

*Proof.* Complete the square,

$$-\tfrac{1}{2}\mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{b}^\top \mathbf{x} = -\tfrac{1}{2}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})^\top \mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) + \tfrac{1}{2}\mathbf{b}^\top \mathbf{A}^{-1}\mathbf{b},$$

shift, and diagonalise $\mathbf{A} = \mathbf{O}\,\mathrm{diag}(\lambda_k)\mathbf{O}^\top$ with $\mathbf{O}$ orthogonal; in the rotated coordinates $\mathbf{y} = \mathbf{O}^\top \mathbf{x}$ the integral factorises into

$$\prod_k \int e^{-\lambda_k y_k^2/2}\, \mathrm{d}y_k = \prod_k \sqrt{2\pi/\lambda_k} = (2\pi)^{L/2}(\det \mathbf{A})^{-1/2}.$$

The density and the generating function follow with $\mathbf{A} = \mathbf{\Sigma}^{-1}$. □

The standard Gaussian on $\mathbb{R}^L$, written $\gamma$ throughout, is the case $\mathbf{\Sigma} = \mathbf{I}$; its coordinates are independent standard Gaussians, which is the fact that reduces every multivariate Hermite statement below to the scalar one.

## 1.2 Isserlis' theorem (Wick's rule)

**Theorem 1.3** (Isserlis)**.** *Let $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$. For any indices $i_1, \dots, i_{2k}$ (repetitions allowed),*

$$\mathbb{E}[X_{i_1} \cdots X_{i_{2k}}] = \sum_{\text{pairings } \pi} \prod_{\{a,b\} \in \pi} \Sigma_{i_a i_b}, \tag{1.3}$$

*the sum over the $(2k-1)!!$ ways of partitioning $\{1, \dots, 2k\}$ into $k$ unordered pairs; products of an odd number of coordinates have expectation zero.*

*Proof sketch.* By Lemma 1.2, $\mathbb{E}[e^{\mathbf{s}^\top \mathbf{X}}] = \exp(\frac{1}{2}\sum_{ab} \Sigma_{ab} s_a s_b)$. The moment $\mathbb{E}[X_{i_1} \cdots X_{i_{2k}}]$ is the mixed derivative $\partial_{s_{i_1}} \cdots \partial_{s_{i_{2k}}}$ at $\mathbf{s} = \mathbf{0}$ of the left side. On the right side, expand the exponential: the term of order $k$ is $\frac{1}{k!}(\frac{1}{2}\sum_{ab} \Sigma_{ab} s_a s_b)^k$, a homogeneous polynomial of degree $2k$ in $\mathbf{s}$; differentiating it $2k$ times and setting $\mathbf{s} = \mathbf{0}$ picks out, for each way of assigning the $2k$ differentiations to the $k$ factors in pairs, the product of the corresponding $\Sigma$ entries. Each unordered pairing arises $k!\,2^k$ times (order of the factors, order within each pair), which cancels the $\frac{1}{k!}\frac{1}{2^k}$ in front. Odd derivatives vanish because the right side is even in $\mathbf{s}$. □

For $k = 2$ this is $\mathbb{E}[X_i X_j X_k X_l] = \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$, the three pairings that appeared in the fourth-cumulant formula of the chapter on moments and cumulants: for a Gaussian vector the fourth moment is exactly the sum over pairings, so the fourth cumulant, which subtracts that sum, vanishes.

## 1.3 Probabilists' Hermite polynomials

**Definition 1.4** (Rodrigues formula)**.** The **probabilists' Hermite polynomial** of degree $n$ is

$$\mathrm{He}_n(x) = (-1)^n e^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2}, \qquad n = 0, 1, 2, \dots \tag{1.4}$$

Differentiating $e^{-x^2/2}$ once gives $-xe^{-x^2/2}$, twice $(x^2-1)e^{-x^2/2}$, and so on; each derivative is a polynomial times $e^{-x^2/2}$, and the sign $(-1)^n$ makes the polynomial monic. The first seven are

$$\begin{aligned} &\mathrm{He}_0(x) = 1, && \mathrm{He}_1(x) = x, && \mathrm{He}_2(x) = x^2 - 1, \\ &\mathrm{He}_3(x) = x^3 - 3x, && \mathrm{He}_4(x) = x^4 - 6x^2 + 3, && \mathrm{He}_5(x) = x^5 - 10x^3 + 15x, \\ &\mathrm{He}_6(x) = x^6 - 15x^4 + 45x^2 - 15. \end{aligned} \tag{1.5}$$

$\mathrm{He}_n$ has the parity of $n$. The coefficient pattern is the Gaussian moment sequence with alternating signs: $\mathrm{He}_n(x) = \sum_k (-1)^k \binom{n}{2k} (2k-1)!!\, x^{n-2k}$, which the generating function below explains.

**Proposition 1.5** (Generating function)**.** *For all real $x$ and $s$,*

$$e^{sx - s^2/2} = \sum_{n \geq 0} \mathrm{He}_n(x) \frac{s^n}{n!}. \tag{1.6}$$

*Proof.* Write $e^{-(x-s)^2/2} = e^{-x^2/2} e^{sx - s^2/2}$. The left side, as a function of $s$, is entire, and its Taylor expansion at $s = 0$ is $\sum_n \frac{s^n}{n!} \frac{\mathrm{d}^n}{\mathrm{d}s^n} e^{-(x-s)^2/2}\big|_{s=0}$. Since $\frac{\mathrm{d}}{\mathrm{d}s} f(x-s) = -f'(x-s)$, the $n$-th derivative at $s = 0$ is $(-1)^n \frac{\mathrm{d}^n}{\mathrm{d}x^n} e^{-x^2/2} = \mathrm{He}_n(x) e^{-x^2/2}$ by Equation (1.4). Dividing by $e^{-x^2/2}$ gives Equation (1.6). □

Two identities follow by differentiating Equation (1.6). Differentiating in $x$, $s\, e^{sx - s^2/2} = \sum_n \mathrm{He}_n'(x) s^n / n!$, and comparing coefficients,

$$\mathrm{He}_n'(x) = n \mathrm{He}_{n-1}(x). \tag{1.7}$$

Differentiating in $s$, $(x - s) e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^{n-1}/(n-1)!$, and comparing coefficients of $s^n/n!$,

$$\mathrm{He}_{n+1}(x) = x \mathrm{He}_n(x) - n \mathrm{He}_{n-1}(x), \tag{1.8}$$

the three-term recursion that `src/markov_score/theory.py` (function `he`) uses to evaluate the polynomials. Expanding $e^{sx} e^{-s^2/2}$ as a product of two series and collecting $s^n$ gives the explicit coefficient formula stated above.

**Proposition 1.6** (Orthogonality)**.** $\mathbb{E}_\gamma[\mathrm{He}_n(Z)\mathrm{He}_m(Z)] = n!\, \delta_{nm}$.

*Proof.* Multiply two copies of Equation (1.6) with parameters $s$ and $u$ and take the expectation under $\gamma$:

$$\mathbb{E}_\gamma\big[e^{sZ - s^2/2} e^{uZ - u^2/2}\big] = e^{-(s^2+u^2)/2} \mathbb{E}_\gamma[e^{(s+u)Z}] = e^{-(s^2+u^2)/2} e^{(s+u)^2/2} = e^{su} = \sum_{n \geq 0} \frac{s^n u^n}{n!}.$$

On the other side the product of the two series is $\sum_{n,m} \mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m]\, s^n u^m / (n!\, m!)$. Comparing the coefficient of $s^n u^m$ gives $\mathbb{E}_\gamma[\mathrm{He}_n \mathrm{He}_m]/(n!\, m!) = \delta_{nm}/n!$, which is the claim. The exchange of expectation and double sum is justified as in Lemma 1.1, the double series being dominated by $e^{(|s|+|u|)|Z|}$. □

So $\{\mathrm{He}_n/\sqrt{n!}\}_{n \geq 0}$ is an orthonormal system in $L^2(\gamma)$, the space of functions $f$ with $\mathbb{E}_\gamma[f(Z)^2] < \infty$. It is also complete: the polynomials are dense in $L^2(\gamma)$ (one proof: if $f \in L^2(\gamma)$ is orthogonal to every polynomial then the entire function $s \mapsto \mathbb{E}_\gamma[f(Z) e^{sZ}]$ has every derivative equal to zero at the origin, hence the Fourier transform of $f\gamma$ vanishes, hence $f = 0$). A function $f \in L^2(\gamma)$ therefore has the **Hermite expansion**

$$f = \sum_{n \geq 0} \frac{\mathbb{E}_\gamma[f \mathrm{He}_n]}{n!} \mathrm{He}_n, \qquad \mathbb{E}_\gamma[f^2] = \sum_{n \geq 0} \frac{\mathbb{E}_\gamma[f \mathrm{He}_n]^2}{n!}, \tag{1.9}$$

with convergence in $L^2(\gamma)$ and the Parseval identity on the right. The **Hermite moment** of degree $n$ of a law $P$ is $\mathbb{E}_P[\mathrm{He}_n]$; when $P$ has a density and $P/\gamma \in L^2(\gamma)$, these are the expansion coefficients of the density ratio: $P/\gamma = \sum_n \mathbb{E}_P[\mathrm{He}_n] \mathrm{He}_n / n!$.

## 1.4 Products of Hermite polynomials and Wick products

**Definition 1.7** (Multivariate Hermite products)**.** For a multi-index $\nu \in \mathbb{N}^L$,

$$H_\nu(\mathbf{x}) = \prod_{i=1}^{L} \mathrm{He}_{\nu_i}(x_i), \qquad |\nu| = \sum_i \nu_i, \qquad \nu! = \prod_i \nu_i!,$$

and $|\nu|$ is the **degree** of $H_\nu$.

Under $\gamma$ on $\mathbb{R}^L$ the coordinates are independent, so the expectation of a product factorises and Proposition 1.6 gives at once

$$\mathbb{E}_\gamma[H_\nu H_\mu] = \prod_i \mathbb{E}_\gamma[\mathrm{He}_{\nu_i}\mathrm{He}_{\mu_i}] = \prod_i \nu_i!\,\delta_{\nu_i\mu_i} = \nu!\,\delta_{\nu\mu}. \tag{1.10}$$

This is the normalisation the paper states in Section 3; $\{H_\nu/\sqrt{\nu!}\}$ is an orthonormal basis of $L^2(\gamma)$ on $\mathbb{R}^L$ (completeness by the same argument coordinatewise), and the **chaos expansion** of $f \in L^2(\gamma)$ is $f = \sum_\nu \mathbb{E}_\gamma[fH_\nu]H_\nu/\nu!$. The paper writes $H_\nu$ with a capital $H$; it is the product of probabilists' polynomials, not a physicists' polynomial (see the chapter on conventions).

The same objects appear in the paper's proofs in a second notation, the **Wick product** $:x_{i_1}\cdots x_{i_n}:$, defined by the multivariate generating function

$$e^{\mathbf{s}^\top\mathbf{x} - |\mathbf{s}|^2/2} = \sum_{n\ge 0} \frac{1}{n!} \sum_{i_1,\dots,i_n} s_{i_1}\cdots s_{i_n}\, :x_{i_1}\cdots x_{i_n}:. \tag{1.11}$$

Since the left side is $\prod_i e^{s_i x_i - s_i^2/2} = \prod_i \sum_k \mathrm{He}_k(x_i)s_i^k/k!$, comparing the coefficient of a monomial $\prod_i s_i^{\nu_i}$ (which appears $n!/\nu!$ times among the ordered tuples $(i_1,\dots,i_n)$) shows

$$:x_{i_1}\cdots x_{i_n}: = H_\nu(\mathbf{x}) \qquad \text{where } \nu_i \text{ is the number of times } i \text{ occurs in } (i_1,\dots,i_n). \tag{1.12}$$

The Wick product is symmetric in its indices and depends only on the multiset. Two degrees are needed explicitly. For degree two, $:x_ix_j: = x_ix_j - \delta_{ij}$, so that $:\mathbf{x}\mathbf{x}^\top: = \mathbf{x}\mathbf{x}^\top - \mathbf{I}$. For degree four, expanding Equation (1.11) or multiplying out $\mathrm{He}_k$'s,

$$\begin{aligned} :x_ix_jx_kx_l: = {} & x_ix_jx_kx_l - \big(\delta_{ij}x_kx_l + \delta_{ik}x_jx_l + \delta_{il}x_jx_k + \delta_{jk}x_ix_l + \delta_{jl}x_ix_k + \delta_{kl}x_ix_j\big) \\ & + \big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big), \end{aligned} \tag{1.13}$$

the inclusion–exclusion over pairs and double pairs; on the diagonal $i=j=k=l$ it gives $x^4 - 6x^2 + 3 = \mathrm{He}_4(x)$, as it must. Equation (1.13) is what the paper means by "the cumulant contractions with $\delta_{ii}$ are exactly the terms that convert monomials into Hermite polynomials".

**Lemma 1.8** (Second moment of a symmetric contraction)**.** *Let* $\mathbf{T}$ *be a symmetric tensor of order* $n$ *on* $\mathbb{R}^L$*. Then*

$$\mathbb{E}_\gamma\Big[\Big(\sum_{i_1\cdots i_n} T_{i_1\cdots i_n}\, :x_{i_1}\cdots x_{i_n}:\Big)^2\Big] = n!\sum_{i_1\cdots i_n} T_{i_1\cdots i_n}^2 = n!\,\|\mathbf{T}\|_F^2. \tag{1.14}$$

*Proof.* Group the ordered tuples by their multiset $\nu$: each $\nu$ with $|\nu| = n$ collects $n!/\nu!$ tuples, all with the same entry $T_\nu$ (symmetry) and the same Wick product $H_\nu$ (Equation (1.12)). So the sum is $\sum_{|\nu|=n}(n!/\nu!)T_\nu H_\nu$, and by Equation (1.10) its second moment is $\sum_\nu (n!/\nu!)^2 T_\nu^2\,\nu! = n!\sum_\nu (n!/\nu!)T_\nu^2 = n!\sum_{\text{ordered}} T^2$. □

*Why it matters here.* This lemma is where the paper's two constants come from. The leading term of the score gap has $m$-th component $\frac{1}{6}\sum_{jkl}\kappa_4[mjkl]\,:x_jx_kx_l:$ times $e^{-4t}$; with $n=3$ and the symmetric tensor $T_{jkl} = \kappa_4[mjkl]$, Equation (1.14) gives a $\gamma$-second moment of $\frac{1}{36}\cdot 3!\sum_{jkl}\kappa_4[mjkl]^2$, and summing over $m$ gives $\frac{1}{6}\|\boldsymbol{\kappa}_4\|_F^2$. The leading term of a shape score is $\frac{1}{24}\sum \partial_\theta\kappa_4[ijkl]\,:x_ix_jx_kx_l:$ times $e^{-4t}$; with $n=4$, its second moment is $\frac{4!}{24^2}\|\partial_\theta\boldsymbol{\kappa}_4\|_F^2 = \frac{1}{24}\|\partial_\theta\boldsymbol{\kappa}_4\|_F^2$. The paper's Appendix A states both computations in one line each; this lemma is that line.

## 1.5 When is a density ratio in $L^2(\gamma)$, and why Theorem 1 does not care

The Hermite expansion Equation (1.9) of a density ratio $P/\gamma$ requires $P/\gamma \in L^2(\gamma)$, that is

$$\int \frac{P(\mathbf{x})^2}{\gamma(\mathbf{x})}\,\mathrm{d}\mathbf{x} < \infty, \qquad \text{equivalently} \qquad \chi^2(P\,\|\,\gamma) = \int \frac{P^2}{\gamma} - 1 < \infty. \tag{1.15}$$

This is a strong condition: it asks $P$ to have tails no heavier than $\sqrt{\gamma}$, that is $P(\mathbf{x}) \lesssim e^{-|\mathbf{x}|^2/4}$ in every direction.

**Proposition 1.9** (Gaussian laws)**.** *Let $P = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$. Then $P/\gamma \in L^2(\gamma)$ if and only if $\lambda_{\max}(\mathbf{\Sigma}) < 2$, and in that case $\int P^2/\gamma = \prod_k \big(\lambda_k(2-\lambda_k)\big)^{-1/2}$ over the eigenvalues $\lambda_k$ of $\mathbf{\Sigma}$.*

*Proof.* $P^2/\gamma = (2\pi)^{-L/2}(\det \mathbf{\Sigma})^{-1}\exp\big(-\frac{1}{2}\mathbf{x}^\top(2\mathbf{\Sigma}^{-1} - \mathbf{I})\mathbf{x}\big)$. By Lemma 1.2 the integral is finite if and only if $2\mathbf{\Sigma}^{-1} - \mathbf{I} \succ 0$, i.e. $\mathbf{\Sigma} \prec 2\mathbf{I}$, and then equals $(\det \mathbf{\Sigma})^{-1}\det(2\mathbf{\Sigma}^{-1} - \mathbf{I})^{-1/2} = \det\big(\mathbf{\Sigma}(2\mathbf{I} - \mathbf{\Sigma})\big)^{-1/2}$. □

Two applications, both stated in the paper's Section 3. First, the clean Gaussian chain at $\rho = 0.85$ has $\lambda_{\max}(\mathbf{\Sigma}_0) > 2$ (the chapter on Markov chains bounds $\lambda_{\max}$ between the row sums of the Toeplitz matrix and $(1+\rho)/(1-\rho)$, and already the $2 \times 2$ block $\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$ has eigenvalue $1 + \rho > 2$ when $\rho > 1$, while for $L = 32$ the Rayleigh quotient of the constant vector exceeds 2 as soon as $\rho \gtrsim 0.35$), so $P_0/\gamma \notin L^2(\gamma)$ for the Gaussian chain itself. Second, the noised law $P_t = \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_t)$ with $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I}$ has eigenvalues $e^{-2t}\lambda_k + 1 - e^{-2t}$, which are all below 2 if and only if $e^{-2t}(\lambda_{\max} - 1) < 1$, i.e.

$$P_t/\gamma \in L^2(\gamma) \iff e^{2t} > \lambda_{\max}(\mathbf{\Sigma}_0) - 1, \tag{1.16}$$

the exact statement in the paper's paragraph "Where the asymptote begins".

**Proposition 1.10** (The Laplace law)**.** *If $P$ is the scalar Laplace density $\frac{1}{2b}e^{-|x|/b}$, then $P/\gamma \notin L^2(\gamma)$ for every $b > 0$. The same holds for the Laplace chain on $\mathbb{R}^L$.*

*Proof.* $P(x)^2/\gamma(x) = \frac{\sqrt{2\pi}}{4b^2}\exp(x^2/2 - 2|x|/b)$, and $x^2/2 - 2|x|/b \to +\infty$, so the integrand is unbounded and not integrable. For the chain, $P_0(\mathbf{a}) = \mu(a_1)\prod_{i\geq 2}\varphi(a_i - \rho a_{i-1})$ with $\varphi$ Laplace; along the ray $\mathbf{a} = u\mathbf{e}_L$ (all sites zero except the last) the ratio $P_0^2/\gamma$ is proportional to $\exp(u^2/2 - 2|u|/b)$, and by continuity the same growth holds on a tube around the ray, so the integral over $\mathbb{R}^L$ diverges. □

Exponential tails are lighter than Gaussian ones in the sense of moments, and the Laplace law has an exponential moment; but they are heavier than $e^{-x^2/4}$, and that is what $L^2(\gamma)$ asks. The same is true of every family in the paper other than the uniform and the mixture.

**Why the paper's Theorem 1 does not need it.** A tempting route to attenuation is spectral: the Ornstein–Uhlenbeck semigroup $(\mathcal{P}_t f)(a) = \mathbb{E}[f(e^{-t}a + \sqrt{\Delta_t}Z)]$ is self-adjoint on $L^2(\gamma)$ with eigenfunctions $\mathrm{He}_n$ and eigenvalues $e^{-nt}$, so one could expand $P_0/\gamma$ in the chaos basis and let the semigroup act coefficient by coefficient. That route needs $P_0/\gamma \in L^2(\gamma)$, which the two propositions above show fails for the paper's chain, Gaussian or Laplace. Theorem 1 is instead an identity for one expectation, $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)]$, proved by Fubini for each fixed $\nu$; it needs only $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$, so that $H_\nu(\mathbf{a})$ is integrable. No expansion of a density is ever summed. This is why the paper can state the theorem for every law with finite moments, and why its proofs of the asymptotic theorems go through the damped cumulant-generating function $\Lambda$ rather than through the chaos expansion of $P_t/\gamma$.

## 1.6 Mehler's formula and the paper's Theorem 1

**Theorem 1.11** (Mehler's formula, moment form)**.** *Let $Z \sim \mathcal{N}(0,1)$, $a \in \mathbb{R}$, $\alpha \in [0,1]$ and $\beta = \sqrt{1-\alpha^2}$. Then for every $n \geq 0$,*

$$\mathbb{E}_Z\big[\mathrm{He}_n(\alpha a + \beta Z)\big] = \alpha^n \mathrm{He}_n(a). \tag{1.17}$$

*Proof.* Put $x = \alpha a + \beta Z$ in the generating function Equation (1.6) and take the expectation over $Z$:

$$\mathbb{E}_Z\big[e^{s(\alpha a+\beta Z)-s^2/2}\big] = e^{s\alpha a - s^2/2}\,\mathbb{E}_Z\big[e^{s\beta Z}\big] = e^{s\alpha a - s^2/2}\,e^{s^2\beta^2/2} = e^{s\alpha a - s^2\alpha^2/2},$$

using Equation (1.1) and $\beta^2 = 1-\alpha^2$. The right-hand side is the generating function Equation (1.6) evaluated at $x = a$ and parameter $\alpha s$: $e^{(\alpha s)a-(\alpha s)^2/2} = \sum_n \mathrm{He}_n(a)\alpha^n s^n/n!$. The left-hand side is $\mathbb{E}_Z[\sum_n \mathrm{He}_n(\alpha a+\beta Z)s^n/n!]$, and the expectation may be taken inside the sum: the series $\sum_n |\mathrm{He}_n(x)||s|^n/n!$ is dominated, for $|s| \leq 1$, by a polynomial in $|x|$ times $e^{|x|}$ (from the explicit coefficient formula, $|\mathrm{He}_n(x)| \leq \sum_k \binom{n}{2k}(2k-1)!!|x|^{n-2k} \leq (1+|x|)^n\, n!!$), which is integrable against the Gaussian law of $Z$, so dominated convergence applies. Comparing the coefficients of $s^n/n!$ on the two sides gives Equation (1.17). □

The name refers to Mehler's 1866 kernel identity,

$$\sum_{n\geq 0} \mathrm{He}_n(x)\mathrm{He}_n(y)\frac{\alpha^n}{n!} = \frac{1}{\sqrt{1-\alpha^2}}\exp\Big(-\frac{\alpha^2(x^2+y^2)-2\alpha xy}{2(1-\alpha^2)}\Big),$$

which is the transition density of the Ornstein–Uhlenbeck process divided by $\gamma(y)$; Equation (1.17) is its moment form, and says that $\mathrm{He}_n$ is an eigenfunction of the channel with eigenvalue $\alpha^n = e^{-nt}$.

**Theorem 1.12** (Hermite attenuation; the paper's Theorem 1)**.** *Let $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ with $\mathbf{a} \sim P_0$ and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ independent. For every multi-index $\nu$ with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$,*

$$\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]. \tag{1.18}$$

*Proof.* Condition on $\mathbf{a}$. Given $\mathbf{a}$, the coordinates $x_{t,i} = e^{-t}a_i + \sqrt{\Delta_t}z_i$ are independent, each of the form $\alpha a_i + \beta z_i$ with $\alpha = e^{-t}$, $\beta = \sqrt{1-\alpha^2} = \sqrt{\Delta_t}$. So

$$\mathbb{E}\big[H_\nu(\mathbf{x}_t)\,\big|\,\mathbf{a}\big] = \prod_i \mathbb{E}_{z_i}\big[\mathrm{He}_{\nu_i}(\alpha a_i + \beta z_i)\big] = \prod_i \alpha^{\nu_i}\mathrm{He}_{\nu_i}(a_i) = \alpha^{|\nu|}H_\nu(\mathbf{a})$$

by Theorem 1.11 applied coordinate by coordinate. Taking the expectation over $\mathbf{a} \sim P_0$ gives Equation (1.18), provided $H_\nu(\mathbf{a})$ is integrable, which holds because $|H_\nu(\mathbf{a})| \leq C(1+|\mathbf{a}|)^{|\nu|}$ and $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$; the tower property $\mathbb{E}[\mathbb{E}[\cdot \mid \mathbf{a}]] = \mathbb{E}[\cdot]$ is Fubini's theorem, which also needs only that integrability. □

The theorem is exact at every $t$, needs no assumption on the dependence between sites, and needs no $L^2(\gamma)$ condition. It is verified by Monte Carlo in `tests/test_theory_hermite.py` on a Laplace sample, and the cases $n \leq 4$ are checked in Lean as polynomial identities under the first four Gaussian moments (`lean/README.md`).

## 1.7 Two exact consequences

**Corollary 1.13** (Cumulants of order three and higher)**.** *For every $m \geq 3$ with $\mathbb{E}|\mathbf{a}|^m < \infty$, the order-$m$ cumulant tensor of $\mathbf{x}_t$ is $e^{-mt}$ times that of $\mathbf{a}$. The covariance is $\mathrm{Cov}(\mathbf{x}_t) = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I}$.*

*Proof.* $\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\mathbf{z}$ is a sum of two independent vectors. By additivity and the transformation rule of the chapter on moments and cumulants, the order-$m$ cumulant tensor of $\mathbf{x}_t$ is $e^{-mt}$ times that of $\mathbf{a}$ plus $\Delta_t^{m/2}$ times that of $\mathbf{z}$; the Gaussian $\mathbf{z}$ has no cumulant tensor of order $m \geq 3$. For $m = 2$ the Gaussian contributes $\Delta_t \mathbf{I}$. □

This corollary uses no Hermite polynomials, only the arithmetic of cumulants; the paper's parenthesis "Gaussian noise has none" is the whole proof. The Hermite form of the theorem is nevertheless the one the paper needs, because Fisher information is built from moments, and the next corollary says how the Hermite moments of degree four see the fourth cumulant.

**Corollary 1.14** (Degree-four Hermite moments and the fourth-cumulant tensor)**.** *Let* $\mathbf{a}$ *have mean zero, covariance* $\mathbf{\Sigma}_0$ *and finite fourth moments. For any indices, with* $\nu$ *the multiset of* $(i, j, k, l)$*,*

$$\mathbb{E}[H_\nu(\mathbf{a})] = \mathbb{E}[{:}a_i a_j a_k a_l{:}] = \kappa_4[ijkl] + (\mathbf{\Sigma}_0 - \mathbf{I})_{ij}(\mathbf{\Sigma}_0 - \mathbf{I})_{kl} + (\mathbf{\Sigma}_0 - \mathbf{I})_{ik}(\mathbf{\Sigma}_0 - \mathbf{I})_{jl} + (\mathbf{\Sigma}_0 - \mathbf{I})_{il}(\mathbf{\Sigma}_0 - \mathbf{I})_{jk}. \tag{1.19}$$

*Consequently, if two laws have the same covariance* $\mathbf{\Sigma}_0$*, their degree-four Hermite moments differ exactly by the difference of their fourth-cumulant tensors, and after noising that difference is* $e^{-4t}(\boldsymbol{\kappa}_4 - \boldsymbol{\kappa}_4')$*.*

*Proof.* Take the expectation of Equation (1.13). By the fourth-cumulant formula of the chapter on moments and cumulants, $\mathbb{E}[a_i a_j a_k a_l] = \kappa_4[ijkl] + \Sigma_{ij}\Sigma_{kl} + \Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk}$ with $\Sigma = \mathbf{\Sigma}_0$; the six pair terms give $-\delta_{ij}\Sigma_{kl} - \dots$; the three double-pair terms give $+\delta_{ij}\delta_{kl} + \dots$. Collecting by pairing, $\Sigma_{ij}\Sigma_{kl} - \delta_{ij}\Sigma_{kl} - \Sigma_{ij}\delta_{kl} + \delta_{ij}\delta_{kl} = (\Sigma - \mathbf{I})_{ij}(\Sigma - \mathbf{I})_{kl}$, and likewise for the other two pairings. The second statement follows because the product terms depend only on $\mathbf{\Sigma}_0$, and the attenuation by $e^{-4t}$ is Theorem 1.12 with $|\nu| = 4$. □

For the paper's variance-matched families, $\mathbf{\Sigma}_0$ is the same for all, so the degree-four Hermite moment of the chain differs from that of its covariance twin by exactly $\boldsymbol{\kappa}_4$; on the diagonal, $\mathbb{E}[\mathrm{He}_4(a_i)] = \kappa_4(a_i)$ because $\mathrm{Var}(a_i) = 1$. Degree two works the same way: $\mathbb{E}[{:}a_i a_j{:}] = (\mathbf{\Sigma}_0 - \mathbf{I})_{ij}$, attenuated by $e^{-2t}$, which is the "correlation coordinate" and is consistent with Corollary 1.13. This is the paper's second exact consequence, checked in Lean as target 2 (`lean/README.md`).

## 1.8 The Edgeworth expansion: the first correction

The paper's E16 asks how much of the gap between the true score and the twin's is explained by "adding the fourth-order term back". The object added is the first term of the Edgeworth (Gram–Charlier A) expansion, defined here; the chapter on misspecification and Edgeworth corrections uses it.

**Scalar form.** Let $X$ have mean 0, variance 1 and cumulants $\kappa_3, \kappa_4, \dots$. Formally, its generating function is $\exp(\frac{1}{2}s^2 + \sum_{n\geq 3} \kappa_n s^n/n!)$, that of the standard Gaussian times $\exp(\sum_{n\geq 3} \kappa_n s^n/n!)$; since multiplying a generating function by $s^n$ corresponds to applying $(-\,\mathrm{d}/\,\mathrm{d}x)^n$ to the density, and $(-\,\mathrm{d}/\,\mathrm{d}x)^n\gamma = \mathrm{He}_n\gamma$ by the Rodrigues formula Equation (1.4),

$$p(x) \text{ “=” } \exp\Big(\sum_{n\geq 3} \frac{\kappa_n}{n!}\big(-\tfrac{\mathrm{d}}{\mathrm{d}x}\big)^n\Big)\gamma(x) = \gamma(x)\Big[1 + \frac{\kappa_3}{6}\mathrm{He}_3(x) + \frac{\kappa_4}{24}\mathrm{He}_4(x) + \dots\Big], \tag{1.20}$$

where the dots collect $\kappa_3^2\mathrm{He}_6/72$ and all higher terms. The **first Edgeworth correction** is the truncation $p \approx \gamma(1 + \kappa_3\mathrm{He}_3/6 + \kappa_4\mathrm{He}_4/24)$, and for a symmetric law it is $p \approx \gamma(1 + \kappa_4\mathrm{He}_4/24)$. The series need not converge (it does not for the Laplace law, whose $\kappa_n$ grow like $n!$), and even when it does the truncation can be negative for large $|x|$; the expansion is an asymptotic device, exact order by order in the cumulants, and the paper reports where it helps and where it diverges rather than assuming either.

**Multivariate form around a Gaussian with covariance $\boldsymbol{\Sigma}$.** Replace $\gamma$ by $\gamma_{\boldsymbol{\Sigma}}$, the density of $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, and the Hermite polynomials by the **$\boldsymbol{\Sigma}$-Hermite tensors**

$$h_{i_1 \cdots i_n}(\mathbf{x}) = (-1)^n \, \frac{\partial_{i_1} \cdots \partial_{i_n} \gamma_{\boldsymbol{\Sigma}}(\mathbf{x})}{\gamma_{\boldsymbol{\Sigma}}(\mathbf{x})}.$$

With $\mathbf{y} = \boldsymbol{\Sigma}^{-1}\mathbf{x}$, $\partial_i \gamma_{\boldsymbol{\Sigma}} = -y_i \gamma_{\boldsymbol{\Sigma}}$ and $\partial_i y_j = (\boldsymbol{\Sigma}^{-1})_{ij}$, so the fourth tensor is

$$\begin{aligned} h_{ijkl}(\mathbf{x}) = {} & y_i y_j y_k y_l - \big[(\boldsymbol{\Sigma}^{-1})_{ij} y_k y_l + \text{five more}\big] \\ & + \big[(\boldsymbol{\Sigma}^{-1})_{ij}(\boldsymbol{\Sigma}^{-1})_{kl} + (\boldsymbol{\Sigma}^{-1})_{ik}(\boldsymbol{\Sigma}^{-1})_{jl} + (\boldsymbol{\Sigma}^{-1})_{il}(\boldsymbol{\Sigma}^{-1})_{jk}\big], \end{aligned} \tag{1.21}$$

Equation (1.13) with $\mathbf{x} \mapsto \mathbf{y}$ and $\delta \mapsto \boldsymbol{\Sigma}^{-1}$. For a symmetric law with fourth-cumulant tensor $\mathbf{K}$ the first correction is $p \approx \gamma_{\boldsymbol{\Sigma}}(1+\epsilon)$ with $\epsilon = \frac{1}{24}\sum_{ijkl} K_{ijkl} h_{ijkl}$; contracting a symmetric $\mathbf{K}$ with the three groups of terms gives, with $A_{kl} = \sum_{ij} K_{ijkl}(\boldsymbol{\Sigma}^{-1})_{ij}$,

$$\epsilon(\mathbf{x}) = \frac{1}{24}\Big[\mathbf{K}[\mathbf{y},\mathbf{y},\mathbf{y},\mathbf{y}] - 6\,\mathbf{y}^\top \mathbf{A}\mathbf{y} + 3\langle \mathbf{A}, \boldsymbol{\Sigma}^{-1}\rangle\Big], \qquad \nabla\epsilon = \frac{1}{6}\boldsymbol{\Sigma}^{-1}\mathbf{K}[\cdot,\mathbf{y},\mathbf{y},\mathbf{y}] - \frac{1}{2}\boldsymbol{\Sigma}^{-1}\mathbf{A}\mathbf{y}, \tag{1.22}$$

where $\mathbf{K}[\mathbf{y},\mathbf{y},\mathbf{y},\mathbf{y}] = \sum K_{ijkl} y_i y_j y_k y_l$ and $\langle \mathbf{A}, \boldsymbol{\Sigma}^{-1}\rangle = \sum_{kl} A_{kl}(\boldsymbol{\Sigma}^{-1})_{kl}$; the six pair terms each contract to $\mathbf{y}^\top\mathbf{A}\mathbf{y}$ and the three double-pair terms each to $\langle \mathbf{A}, \boldsymbol{\Sigma}^{-1}\rangle$ by the symmetry of $\mathbf{K}$, and the gradient uses $\nabla_{\mathbf{x}} = \boldsymbol{\Sigma}^{-1}\nabla_{\mathbf{y}}$. The score of the corrected density is $\nabla \log[\gamma_{\boldsymbol{\Sigma}}(1+\epsilon)] = -\boldsymbol{\Sigma}^{-1}\mathbf{x} + \nabla\epsilon/(1+\epsilon)$. This is exactly the formula in the paper's Appendix E for E16, with $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_t$ and $\mathbf{K} = e^{-4t}\boldsymbol{\kappa}_4$: by Corollary 1.13 the noised chain has covariance $\boldsymbol{\Sigma}_t$ and fourth-cumulant tensor $e^{-4t}\boldsymbol{\kappa}_4$, so the first Edgeworth correction around the covariance twin is the degree-four Hermite term that the paper's Theorem 2 identifies as the whole of the leading gap.

## 1.9 Worked example: a unit-variance Laplace site at $t = \log 2$

Let $a$ have the unit-variance Laplace law, $b = 1/\sqrt{2}$, with $\kappa_4 = 3$ (chapter on moments and cumulants), and let $x_t = e^{-t}a + \sqrt{\Delta_t}z$ at $t = \log 2$, so $\alpha = e^{-t} = \frac{1}{2}$ and $\Delta_t = \frac{3}{4}$. Theorem 1 predicts

$$\mathbb{E}[\mathrm{He}_4(x_t)] = \alpha^4 \mathbb{E}[\mathrm{He}_4(a)] = \tfrac{1}{16}(\mathbb{E}a^4 - 6\mathbb{E}a^2 + 3) = \tfrac{1}{16}(6 - 6 + 3) = \tfrac{3}{16},$$

using $\mathbb{E}a^4 = 24b^4 = 6$. Directly: $\mathbb{E}x_t^2 = \frac{1}{4} + \frac{3}{4} = 1$ and $\mathbb{E}x_t^4 = \alpha^4\mathbb{E}a^4 + 6\alpha^2\Delta_t\mathbb{E}a^2\mathbb{E}z^2 + \Delta_t^2\mathbb{E}z^4 = \frac{6}{16} + 6\cdot\frac{1}{4}\cdot\frac{3}{4} + \frac{9}{16}\cdot 3 = \frac{6+18+27}{16} = \frac{51}{16}$, so $\mathbb{E}[\mathrm{He}_4(x_t)] = \frac{51}{16} - 6 + 3 = \frac{3}{16}$, as predicted. The fourth cumulant of $x_t$ is $\kappa_4(x_t) = \mathbb{E}x_t^4 - 3 = \frac{3}{16} = e^{-4t}\kappa_4(a)$, illustrating Corollary 1.13; and the degree-two moment $\mathbb{E}[\mathrm{He}_2(x_t)] = \mathbb{E}x_t^2 - 1 = 0$ equals $e^{-2t}\mathbb{E}[\mathrm{He}_2(a)] = 0$, as unit variance requires.

For the constant of Lemma 1.8 in the scalar case: the leading term of the score gap is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$, whose $\gamma$-second moment is $(\kappa_4/6)^2 \cdot 3!\, e^{-8t} = (\kappa_4^2/6)e^{-8t}$, that is $\frac{9}{6}e^{-8t} = 1.5\, e^{-8t}$ for the Laplace law; this is the value that `tests/test_theory_hermite.py` compares with a fine quadrature of the scalar noised density.

*Where it is used.* The paper's Section 3 (Theorem 1 and its two consequences, the Hermite normalisation $\nu!$, the $L^2(\gamma)$ remarks, the Edgeworth paragraph "The gap is the fourth-order term"), Appendix A (the proofs of Theorems 1 and 2, Wick's theorem for the constants, the crossover $e^{2t} > \lambda_{\max} - 1$) and Appendix E (E16). Experiments E01, E16, E18, E21. Code: the functions `he`, `hermite_moment_attenuation`, `score_gap_constant` and `shape_information_constant` of `src/markov_score/theory.py`; the Edgeworth oracle `experiments/e16_edgeworth_oracle.py`; the tests in `tests/test_theory_hermite.py`; and the Lean targets listed in `lean/README.md`.

**References.** G. Szegő (1975), *Orthogonal Polynomials*, 4th edition, American Mathematical Society (Chapter V: Hermite polynomials, generating function, Mehler's formula). L. Isserlis (1918), "On a formula for the product-moment coefficient of any order of a normal frequency

distribution in any number of variables", Biometrika 12, 134–139. F. G. Mehler (1866), "Ueber die Entwicklung einer Function von beliebig vielen Variablen nach Laplaceschen Functionen höherer Ordnung", Journal für die reine und angewandte Mathematik 66, 161–176. S. Janson (1997), *Gaussian Hilbert Spaces*, Cambridge University Press (Wick products, chaos expansion, the Ornstein–Uhlenbeck semigroup on $L^2(\gamma)$). P. Hall (1992), *The Bootstrap and Edgeworth Expansion*, Springer (Chapter 2: the Edgeworth expansion and its first terms). P. McCullagh (1987), *Tensor Methods in Statistics*, Chapman and Hall (Chapter 5: Edgeworth series in several variables, Hermite tensors).